

FLATRUET, Sweden

WMO: 023030

Lat: 62.7291N

Lon: 12.7264E

Elev: 3058

StdP: 13.14

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-1.1	-11.5	3.3	0.5	-7.6	4.1	4.0	47.1	26.3	43.1	25.3	4.6	0	0.867

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	68.3	54.7	65.1	53.2	61.7	52.0	57.4	63.7	55.7	61.4	53.9	59.2	9.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.8	71.7	58.6	53.0	66.9	57.2	51.5	63.2	56.0	25.9	64.0	24.8	61.1	23.7	59.4	63.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.5	30.1	25.6	DB	-9.2	73.0	5.6	2.6	-13.3	74.9	-16.6	76.4	-19.7	77.8	-23.8	79.7	(4)
			WB	-10.0	59.3	4.7	2.8	-13.4	61.3	-16.2	62.9	-18.8	64.5	-22.3	66.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	17.5	17.6	21.5	29.5	37.3	46.2	51.3	50.5	42.0	32.7	24.9	19.3
	DBStd	14.23	8.41	8.62	7.94	6.76	6.88	6.96	5.47	5.86	6.03	7.14	7.94	8.61
	HDD50	6577	1008	909	883	614	400	155	48	66	249	538	754	953
	HDD65	11825	1473	1329	1348	1064	859	563	425	449	691	1002	1204	1418
	CDD50	228	0	0	0	0	6	43	89	82	8	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	3	0	0	0	0	0	2	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.4	12.1	12.0	13.7	12.3	11.6	11.5	10.2	9.4	11.2	11.4	10.9	10.5
Precipitation	PrecAvg	24.6	1.9	1.4	1.3	1.3	1.9	2.8	3.4	3.2	2.0	2.0	1.9	1.5
	PrecMax	30.9	4.0	2.3	1.9	3.1	3.7	5.2	6.6	5.4	3.6	4.6	3.8	2.3
	PrecMin	20.0	0.3	0.2	0.4	0.3	0.6	0.9	1.0	1.5	0.7	0.8	0.7	0.6
	PrecStd	3.0	0.9	0.6	0.4	0.6	0.9	1.2	1.4	1.1	0.9	1.0	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	36.9	39.7	48.5	65.1	71.2	72.3	70.2	61.8	50.7	44.1	37.3
		MCWB	31.5	30.6	33.2	40.8	52.5	54.3	58.7	55.3	49.7	45.5	38.1	34.3
	2%	DB	32.7	32.8	36.3	44.1	57.9	67.8	68.1	66.5	57.8	47.0	40.9	35.2
		MCWB	30.7	30.6	33.1	36.5	47.0	52.8	56.3	53.3	48.4	42.7	37.7	33.1
	5%	DB	30.8	30.9	34.1	41.4	51.8	63.2	64.5	63.5	53.8	44.4	37.4	33.0
		MCWB	29.4	29.2	31.3	35.5	44.1	49.9	55.1	53.0	47.8	41.1	35.4	31.2
	10%	DB	28.6	28.6	32.0	38.6	47.5	58.3	61.0	60.2	50.9	42.2	35.0	30.6
		MCWB	27.3	26.9	29.5	34.2	41.0	48.5	52.9	52.5	46.4	39.5	33.5	29.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	33.5	35.5	41.9	53.5	57.1	59.8	59.7	52.9	47.5	41.2	35.7
		MCDB	33.5	34.5	36.9	46.7	62.7	63.7	68.9	65.4	57.2	49.3	42.7	36.3
	2%	WB	31.4	31.0	33.9	38.5	48.8	54.8	57.6	57.3	50.6	43.9	38.3	33.3
		MCDB	32.2	31.9	35.8	42.5	56.5	63.6	65.2	62.3	54.6	45.9	39.9	34.6
	5%	WB	29.6	29.2	31.9	36.6	45.1	52.5	56.2	55.2	48.9	41.8	35.8	31.5
		MCDB	30.6	30.6	33.9	39.8	50.2	60.3	62.2	59.7	52.7	43.7	36.9	32.7
	10%	WB	27.4	27.0	29.7	34.9	42.2	49.8	54.5	53.4	46.8	40.0	33.4	29.1
		MCDB	28.5	28.8	31.4	37.8	46.7	57.3	59.5	57.4	49.9	41.9	34.5	30.4
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.6	9.2	10.2	10.9	13.3	12.5	12.0	10.5	8.0	7.3	8.4
		MCDBR	8.1	9.0	9.5	12.6	17.0	19.8	17.5	16.4	15.0	11.2	8.9	10.0
		MCWBR	7.9	8.5	7.7	7.8	9.6	9.5	8.7	7.8	8.4	8.0	7.8	9.8
	5% WB	MCDBR	8.0	8.8	9.4	11.5	14.9	17.5	16.2	13.6	13.4	10.5	8.8	9.6
		MCWBR	8.1	8.9	8.4	7.8	9.6	9.5	9.2	8.1	8.7	8.2	8.3	9.6
Clear-Sky Solar Irradiance	taub	0.187	0.243	0.271	0.302	0.313	0.311	0.323	0.312	0.286	0.269	0.198	0.166	
	taud	2.053	2.255	2.269	2.319	2.473	2.529	2.502	2.519	2.580	2.482	2.117	1.964	
	Ebn at Noon	161	233	268	282	288	291	283	277	266	225	153	99	
	Edh at Noon	10	18	27	32	30	29	29	26	20	15	9	5	
All-Sky Solar Radiation	RadAvg	68	254	713	1301	1499	1593	1479	1137	711	308	93	34	
	RadStd	6	39	64	107	108	139	174	106	75	42	15	3	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A	

Nomenclature: See separate page